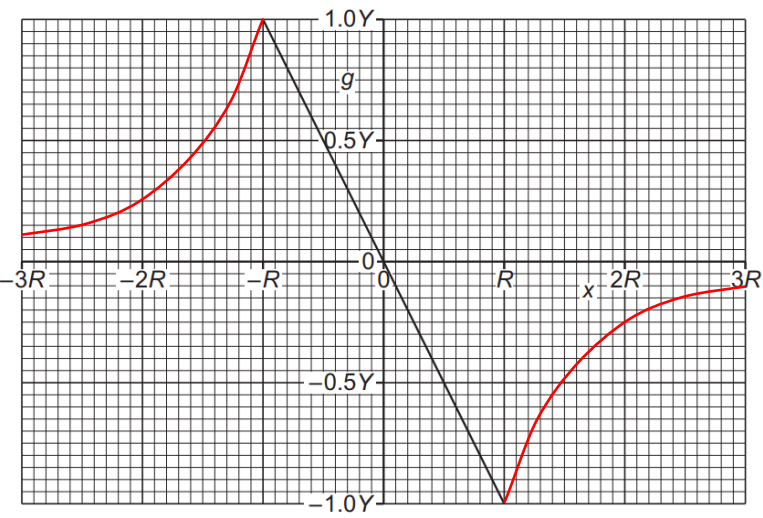


2(a)	The <u>change in height is negligible compared with radius</u> of the planet. Thus, the field strength $g = \frac{GM}{R^2}$ remains relatively constant for any small change in R. The gravitational field lines are effectively parallel.	[1]
2(b)(i)	$Y = \frac{GM}{R^2}$ where G is the gravitational constant	[1]

2(b)(ii)		[1] for correct shape of curve from $-3R$ to $-R$; ending at $(-R, 1.0Y)$ [1] for correct shape of curve from R to $3R$; starting at $(R, -1.0Y)$ [1] for curves passing through $(-3R, 0.11Y)$; $(-2R, 0.25Y)$; $(2R, -0.25Y)$; $(3R, -0.11Y)$ (don't need to mark for $\pm 3R$)
2(b)(iii)	Initial total energy = Final total energy $\frac{1}{2}mv^2 - \frac{GMm}{R} = \frac{1}{2}\left(\frac{1}{2}mv^2\right) - \frac{GMm}{x}$ $\frac{1}{4}v^2 = GM\left(\frac{1}{R} - \frac{1}{x}\right)$ $\frac{1}{4}(4.7 \times 10^3)^2 = (6.67 \times 10^{-11})(6.4 \times 10^{23})\left(\frac{1}{3.4 \times 10^6} - \frac{1}{x}\right)$ $x = 6.070 \times 10^6 \text{ m}$ Therefore distance travelled $= x - R$ $= 6.070 \times 10^6 - 3.4 \times 10^6$ $= 2.670 \times 10^6 \text{ m}$ $= 2.7 \times 10^6 \text{ m}$	[1] for correct equation for conservation of energy [1] for correct substitution [1] for correct answer
2(c)	The electric field strength in both sphere A (between $x = 0$ and	[1]